Yiqing Xie

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Education

Carnegie Mellon University 2022 – 2027 (Expected)

- Ph.D. in Language and Information Technology
- Advisors: Carolyn Rosé and Daniel Fried
- Thesis: Synthetic Training Data Construction for Coding Agents
  Committee: Carolyn Rosé, Daniel Fried, Graham Neubig, and Dan Roth (UPenn)

University of Illinois Urbana-Champaign 2020 – 2022

- M.S. in Computer Science (GPA: 4.0/4.0)
- Advisor: Jiawei Han

Hong Kong University of Science and Technology 2016 – 2020

- B.Sc. in Computer Science; double major in Mathematics (GPA: 3.9/4.3)

Research Highlights

- **Agentic Post-Training**. I construct scalable and effective agent training environments [COLM 2025], design lightweight training tasks that generalize to multiple coding agent benchmarks [ICML 2026], and tackle reward-hacking for rubric-based RL [preprint].
- **Coding Agents**. I develop multi-agent inference settings that are effective, flexible, and cost-efficient [EMNLP 2026], design automatic analysis methods that efficiently discover the influential factors of coding agent training demonstrations [preprint], and build evaluation benchmarks for realistic workplace tasks [Neurips 2025].
- **Factuality and Reliability**. I train critique-based evaluators that substantially improve factuality training [ACL 2025], design evaluation frameworks for reliable and trustworthy clinical text generation [ACL 2024], and continuous pretrain LLMs for better generalizability [SIGIR 2023].

Industrial Experiences

Meta AI May 2024 – Oct. 2024

- Research Intern, GenAI
- Manager: Hejia Zhang; Peers: Di Jin, Pradyot Prakash, and Sinong Wang.
- Trained a fine-grained critique-based evaluator and used it to improve generators' factuality (ACL 2025).

Microsoft Research, Redmond June 2023 – Aug. 2023

- Research Intern, Health Futures
- Managers: Sheng Zhang, Hao Cheng, and Hoifung Poon.
- Developed a multi-aspect, fine-grained evaluation framework for medical text generation (ACL 2024).

Microsoft Research, Redmond May 2022 – Aug. 2022

- Research Intern, Productivity and Intelligence Group
- Manager: Chenyan Xiong.
- Worked on continuously pretrained models for zero-shot dense retrieval (SIGIR 2023).

Alibaba DAMO Academy July 2020 – Feb. 2021

- Research Intern, Data Analytics and Intelligence Lab
- Managers: Yaliang Li and Bolin Ding.
- Worked on few-shot interaction recommendation across multiple scenarios (WWW 2022).

Selected Publications

For the full list, please refer to scholar.google.com/citations?user=ov9EcNEAAAAJ. * denotes equal contribution.

Successful Rollouts are Not Equal: Discovering Influential Behaviors for Training Coding Agents Preprint, 2026
- **Yiqing Xie**, Gaokai Zhang, Xingyao Wang, Dan Roth, Daniel Fried, Carolyn Rose

Eat Your Precision: Precision Objective Mitigates Reward Hacking in Rubric-based RL Preprint, 2026
- Hyun Ryu, Jong-Heon Kim, **Yiqing Xie**, Seungone Kim, Carolyn Rose, Eunho Yang

Steer, Don't Solve: Training Small Critic Models for Large Code Agents EMNLP, 2026
- Shubham Gandhi*, **Yiqing Xie*** , Atharva Naik, Ruichen Zhu, and Carolyn Rose

Mind the Sim2Real Gap in User Simulation for Agentic Tasks COLM, 2026
- Xuhui Zhou, Weiwei Sun, Qianou Ma, **Yiqing Xie**, Jiarui Liu, Weihua Du, Sean Welleck, Yiming Yang, Graham Neubig, Sherry Tongshuang Wu, and Maarten Sap

Hybrid-Gym: Training Coding Agents to Generalize Across Tasks ICML, 2026
- **Yiqing Xie**, Emmy Liu, Gaokai Zhang, Nachiket Kotalwar, Shubham Gandhi, Sathwik Acharya, Xingyao Wang, Carolyn Rose, Graham Neubig, and Daniel Fried

RepoST: Scalable Repository-Level Coding Environment Construction with Sandbox Testing COLM, 2025
- **Yiqing Xie**, Alex Xie, Divyanshu Sheth, Pengfei Liu, Daniel Fried, and Carolyn Rose

TheAgentCompany: Benchmarking LLM Agents on Consequential Real World Tasks NeurIPS, 2025
- Frank F. Xu, Yufan Song, Boxuan Li, Yuxuan Tang, Kritanjali Jain, Mengxue Bao, Zora Z. Wang, Xuhui Zhou, Zhitong Guo, Murong Cao, Mingyang Yang, Hao Yang Lu, Amaad Martin, Zhe Su, Leander Maben, Raj Mehta, Wayne Chi, Lawrence Jang, **Yiqing Xie**, Shuyan Zhou, and Graham Neubig

Improving Model Factuality with Fine-grained Critique-based Evaluator ACL, 2025
- **Yiqing Xie**, Wenxuan Zhou, Pradyot Prakash, Di Jin, Yuning Mao, Quintin Fettes, Arya Talebzadeh, Sinong Wang, Han Fang, Carolyn Rose, Daniel Fried, and Hejia Zhang

CodeRAG-Bench: Can Retrieval Augment Code Generation? NAACL Findings, 2025
- Zora Zhiruo Wang, Akari Asai, Xinyan Velocity Yu, Frank F. Xu, **Yiqing Xie**, Graham Neubig, and Daniel Fried

DocLens: Multi-aspect Fine-grained Evaluation for Medical Text Generation ACL, 2024
- **Yiqing Xie**, Sheng Zhang, Hao Cheng, Pengfei Liu, Zelalem Gero, Cliff Wong, Tristan Naumann, Hoifung Poon, and Carolyn Rose

Data Augmentation for Code Translation with Comparable Corpora and Multiple References EMNLP Findings, 2023
- **Yiqing Xie**, Atharva Naik, Daniel Fried, and Carolyn Rose

Unsupervised Dense Retrieval Training with Web Anchors SIGIR, 2023
- **Yiqing Xie**, Xiao Liu, and Chenyan Xiong

Invited Talks

Designing Scalable and Generalizable Synthetic Training Tasks for Coding Agents
- Nvidia Reading Club, Nvidia — June 2026

Synthetic Data Approaches for Coding Agent Training
- Guest Lecture for 11-891, Neural code generation, Carnegie Mellon University — Nov 2025

Constructing Scalable Repository-Level Environments for Coding Agent Training
- EFML Reading Group, Stanford / University of Washington — Apr 2025

Multi-aspect Fine-grained Evaluation for Medical Text Generation
- Health and Life Science Seminar, Microsoft Research — Apr 2024

Student Mentoring

Shubham Gandhi Sept. 2024 – Present
- CMU MIIS Master's Student → CMU Ph.D. Student

Dhruv Gupta Dec. 2024 – May 2025
- CMU MIIS Master's Student → Skild AI

Gayathri Ganesh Lakshmy Dec. 2024 – May 2025
- CMU MIIS Master's Student → Databricks

Gaokai Zhang Nov. 2025 – Present
- CMU MIIS Master's Student

Ruichen Zhu May 2026 – Aug. 2026
- CMU Undergraduate Student

Hyun Ryu Aug. 2026 – Oct. 2026
- KAIST Master's Student

Jerry Yan Sept. 2026 – Present
- CMU MCDS Master's Student

Honors and Awards

- CMU Presidential Fellowship in LTI 2024 – 2025
- Siebel Scholar, Class of 2022 2021 – 2022
- HKUST Academic Achievement Medal (top 1%) 2020
- HKSAR Government Scholarship Fund – Reaching Out Award 2018
- HKUST Scholarship for Continuing Undergraduate Students 2017 – 2019
- Dean's List, HKUST (three times) 2017 – 2019